

Transmit Beampattern Optimization Algorithm in Networked Dual-function Radar-communication System for Low Probability of Interception

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Abstract—In the scenario of the multi-aircraft networked dual-function radar-communication (DFRC) system, the traditional beamforming merely focuses on satisfying the radar sensing and communication performance, without considering the overall radio frequency radiation management, thereby degrading its low probability of interception (LPI) performance. To solve the above problem, a transmit beampattern optimization algorithm in the networked DFRC system for LPI performance is proposed. With the optimization objective of minimizing the total radiation power by optimizing the transmit beamforming vector and radiation power of each aircraft, a transmit beampattern optimization model for the networked DFRC system is formulated under the constraints of the predefined radar signal to interference plus noise ratio (SINR) and communication SINR, thereby improving the LPI performance of the underlying system. To address the non-convexity of the optimization problem, an iterative solution technique is proposed by combining the alternating optimization (AO) method with the successive convex approximation (SCA) approach. Simulation results validate that the proposed algorithm can provide lower radiation power consumption while satisfying the predetermined radar and communication performance demands, thereby enhancing the LPI performance of the networked DFRC system.

Index Terms—Dual-function radar-communication (DFRC), low probability of interception (LPI), transmit beampattern, radar sensing, communication transmission, signal to interference plus noise ratio (SINR).

I. INTRODUCTION

In modern electronic warfare and information-centric conflict environments, dual-function radar-communication (DFRC) technology breaks the limitation of the separate deployment of traditional radar and communication equipment by sharing the radio frequency front-end, spectrum resources, and antenna array [1]. Networked DFRC technology further endows the system with spatial diversity and cooperative gains, which can not only improve the target resolution accuracy through multi-node joint detection but also realize reliable communication between nodes based on distributed links [2].

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As the core support of the DFRC system, beamforming technology realizes directional energy allocation by adjusting the amplitude and phase weight of the antenna array. It is a key technology for balancing detection performance, communication quality, and low probability of interception (LPI) performance. The LPI performance of the networked DFRC system needs to ensure the collaborative detection of multiple nodes and distributed communication. At the same time, the leakage of radiation energy must be suppressed to avoid interception and localization by the passive detection system of the enemy [3] [4] [5].

Currently, extensive research efforts have been carried out by scholars worldwide in this area. In single-platform, frequency diverse array multiple-input multiple-output (FDA-MIMO) [6] and time-modulated array (TMA) techniques [7] have demonstrated promising performance. Corresponding beamforming algorithms typically rely on weighted minimum mean square error (WMMSE) [8], alternating direction method of multipliers (ADMM) [9], or related iterative optimization frameworks to achieve beam pattern modulation [10]. However, it is difficult to directly transfer to a networked cluster, and it does not consider the requirements of radar detection, communication, and LPI performance, and most of the links are assumed to be ideal symmetrical, which is not consistent with the actual cluster channel heterogeneity scenario.

Therefore, this paper proposes a transmit beampattern optimization algorithm to improve the LPI performance of the networked DFRC system. Firstly, the analytical expressions for the radar signal to interference plus noise ratio (SINR) and communication SINR are derived, respectively. Subsequently, a transmit beampattern optimization model is constructed to minimize the total radiation power of the networked DFRC system, while adhering to the constraints of meeting the preset radar SINR and communication SINR thresholds. To effectively solve the non-convex optimization problem in the optimization model, an iterative solution strategy is proposed by combining the alternating optimization (AO) method with the successive convex approximation (SCA) method. Simulation results validate that the proposed algorithm can provide lower radiation power while satisfying the above radar and

communication performance, thereby facilitating the LPI performance of the networked DFRC system.

II. SYSTEM MODEL

A. Networked DFRC System Model

Consider a scenario of the networked DFRC system consisting of N aircraft and T targets, as illustrated in Fig. 1. Each aircraft is equipped with a uniform linear array of antennas with a total of $M = 16$ antennas. The aircraft perform the task of detecting targets as well as the task of communicating with each other.

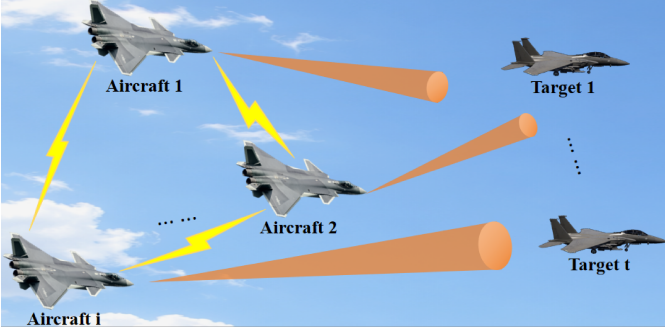


Fig. 1. Networked DFRC System Model

The transmitting signal of the aircraft i is expressed as,

$$\mathbf{x}_i = \sqrt{p_i} \mathbf{w}_i \mathbf{s}_i \quad (1)$$

where p_i denotes the total radiation power of the aircraft i , $\mathbf{w}_i$ denotes the transmit beamforming vector, and $\mathbf{s}_i$ is the DFRC complex baseband signal, which satisfies the energy normalization constraint $\mathbb{E}[|\mathbf{s}_i|^2] = 1$.

B. Radar Performance Metric

Assuming the target is at angle $\theta_{i,t}$, the received radar echo from target t can be expressed as,

$$\mathbf{y}_{i,t}^{\text{rad}} = \alpha_{i,t} \mathbf{a}_r(\theta_{i,t}) \mathbf{a}_t^H(\theta_{i,t}) \mathbf{x}_i + \sum_{\substack{k=1 \\ k \neq i}}^N \alpha_{k,t} \mathbf{a}_r(\theta_{i,t}) \mathbf{a}_t^H(\theta_{i,t}) \mathbf{x}_k + \mathbf{n}_r \quad (2)$$

where α denotes the target reflection coefficient, $\mathbf{a}_r(\theta) = \mathbf{a}_t(\theta) = [1, e^{j2\pi d \sin(\theta)/\lambda}, \dots, e^{j2\pi(M-1)d \sin(\theta)/\lambda}]^T$ denote the transmit and receive steering vectors respectively, d and λ denote the array element spacing and signal wavelength respectively, and $\mathbf{n}_r$ denotes the complex white Gaussian noise.

To extract the radar echo and suppress the interference, the received radar echo can be rewritten as,

$$\hat{\mathbf{y}}_{i,t}^{\text{rad}} = \mathbf{u}_{i,t}^H \mathbf{y}_{i,t}^{\text{rad}} \quad (3)$$

where $\mathbf{u}_i$ denotes the radar receiving beamforming vector.

For the radar task, the radar detection SINR can be expressed as,

$$\text{SINR}_{i,t}^{\text{rad}} = \frac{p_i \alpha_{i,t}^2 |\mathbf{u}_{i,t}^H \mathbf{a}_r(\theta_{i,t})|^2 |\mathbf{a}_t^H(\theta_{i,t}) \mathbf{w}_i|^2}{\sum_{\substack{k=1 \\ k \neq i}}^N p_k \alpha_{k,t}^2 |\mathbf{u}_{i,t}^H \mathbf{a}_r(\theta_{i,t})|^2 |\mathbf{a}_t^H(\theta_{i,t}) \mathbf{w}_k|^2 + \sigma^2 \mathbf{u}_{i,t}^H \mathbf{u}_{i,t}} \quad (4)$$

where σ^2 denotes the noise power.

C. Communication Performance Metric

The communication received signal from aircraft i to aircraft j can be expressed as,

$$\mathbf{y}_{i,j}^{\text{com}} = \mathbf{h}_{i,j} \mathbf{x}_i + \sum_{\substack{k=1 \\ k \neq i}}^N \mathbf{h}_{k,j} \mathbf{x}_k + \mathbf{n}_c \quad (5)$$

where $\mathbf{h}_{i,j}$ denotes the communication channel matrix from aircraft i to j , and $\mathbf{n}_c$ denotes the additive complex white Gaussian noise.

For the communication task, the received communication SINR can be expressed as,

$$\text{SINR}_{i,j}^{\text{com}} = \frac{p_i |\mathbf{h}_{i,j}^H \mathbf{w}_i|^2}{\sum_{\substack{k=1 \\ k \neq i}}^N p_k |\mathbf{h}_{k,j}^H \mathbf{w}_k|^2 + \sigma^2} \quad (6)$$

where σ^2 denotes the noise power.

It is important to note that the communication links are not required to lie strictly within the mainlobe of the transmit beam. Therefore, as long as the communication SINR constraint is satisfied, communication links can still be reliably supported through sidelobe radiation. Consequently, both radar targets and communication tasks can be simultaneously served under a unified beamforming framework.

III. TRANSMIT BEAMPATTERN OPTIMIZATION ALGORITHM

A. Problem Formulation

In this paper, we consider a networked DFRC system and aim to minimize the total radiation power while satisfying both radar sensing and communication performance requirements. The optimization model is established as follows,

$$\begin{aligned} \min_{\mathbf{w}_i, p_i} \quad & \sum_{i=1}^N p_i \\ \text{s.t.} \quad & \begin{cases} \text{SINR}_{i,t}^{\text{rad}} \geq \gamma_r \\ \text{SINR}_{i,j}^{\text{com}} \geq \gamma_c \\ P_{\min} \leq p_i \leq P_{\max} \\ \|\mathbf{w}_i\|_2 = 1 \end{cases} \end{aligned} \quad (7)$$

where γ_r and γ_c denote the radar SINR and communication SINR thresholds, respectively.

It is assumed that the radar receiving vector is given by,

$$\mathbf{u}_{i,t} = \frac{\mathbf{a}_r(\theta_{i,t})}{\|\mathbf{a}_r(\theta_{i,t})\|_2} \quad (8)$$

which can be seen as a known constant. Consider a monostatic radar configuration where the transmit and receive arrays share the same steering vector, $\mathbf{a}_t(\theta_{i,t}) = \mathbf{a}_r(\theta_{i,t}) \triangleq \mathbf{a}(\theta_{i,t})$. Substituting (8) into (4), the radar SINR can be rewritten as,

$$\text{SINR}_{i,t}^{\text{rad}} = \frac{p_i \alpha_{i,t}^2 |\mathbf{a}(\theta_{i,t})|^2 |\mathbf{a}^H(\theta_{i,t}) \mathbf{w}_i|^2}{\sum_{k=1, k \neq i}^N p_k \alpha_{k,t}^2 |\mathbf{a}(\theta_{i,t})|^2 |\mathbf{a}^H(\theta_{i,t}) \mathbf{w}_k|^2 + \sigma^2} \quad (9)$$

The power minimization problem is formulated as,

$$\begin{aligned} \min_{\mathbf{w}_i, p_i} \quad & \sum_{i=1}^N p_i \\ \text{s.t.} \quad & \begin{cases} \frac{p_i \alpha_{i,t}^2 |\mathbf{a}(\theta_{i,t})|^2 |\mathbf{a}^H(\theta_{i,t}) \mathbf{w}_i|^2}{\sum_{k=1, k \neq i}^N p_k \alpha_{k,t}^2 |\mathbf{a}(\theta_{i,t})|^2 |\mathbf{a}^H(\theta_{i,t}) \mathbf{w}_k|^2 + \sigma^2} \geq \gamma_r \\ \frac{p_i |\mathbf{h}_{i,j}^H \mathbf{w}_i|^2}{\sum_{k=1, k \neq i}^N p_k |\mathbf{h}_{k,j}^H \mathbf{w}_k|^2 + \sigma^2} \geq \gamma_c \\ 0 \leq p_i \leq P_{\max} \\ \|\mathbf{w}_i\|_2 = 1 \end{cases} \end{aligned} \quad (10)$$

B. Solution Technique

Problem (10) is non-convex due to the coupled variables $(p_i, \mathbf{w}_i)$ and the quadratic magnitude terms. We adopt an AO method framework combined with the SCA method to obtain a stationary solution.

Subproblem 1: Power Optimization with Fixed Beamforming. The radar sensing constraint yields a linear lower bound on p_i , that is,

$$p_i \geq \frac{\gamma_r (\sum_{k=1, k \neq i}^N p_k \alpha_{k,t}^2 |\mathbf{a}(\theta_{i,t})|^2 |\mathbf{a}^H(\theta_{i,t}) \mathbf{w}_k|^2 + \sigma^2)}{\alpha_{i,t}^2 |\mathbf{a}(\theta_{i,t})|^2 |\mathbf{a}^H(\theta_{i,t}) \mathbf{w}_i|^2} \quad (11)$$

Similarly, the received communication SINR constraint can be rewritten as,

$$p_i \geq \frac{\gamma_c (\sum_{k=1, k \neq i}^N p_k |\mathbf{h}_{k,j}^H \mathbf{w}_k|^2 + \sigma^2)}{|\mathbf{h}_{i,j}^H \mathbf{w}_i|^2} \quad (12)$$

Thus, the power optimization subproblem becomes a linear program, which can be shown as follows,

$$\begin{aligned} \min_{p_i} \quad & \sum_{i=1}^N p_i \\ \text{s.t.} \quad & \begin{cases} p_i \geq \frac{\gamma_r (\sum_{k=1, k \neq i}^N p_k \alpha_{k,t}^2 |\mathbf{a}(\theta_{i,t})|^2 |\mathbf{a}^H(\theta_{i,t}) \mathbf{w}_k|^2 + \sigma^2)}{\alpha_{i,t}^2 |\mathbf{a}(\theta_{i,t})|^2 |\mathbf{a}^H(\theta_{i,t}) \mathbf{w}_i|^2} \\ p_i \geq \frac{\gamma_c (\sum_{k=1, k \neq i}^N p_k |\mathbf{h}_{k,j}^H \mathbf{w}_k|^2 + \sigma^2)}{|\mathbf{h}_{i,j}^H \mathbf{w}_i|^2} \\ 0 \leq p_i \leq P_{\max} \end{cases} \end{aligned} \quad (13)$$

Algorithm 1 AO-SCA for Solving (10)

- 1: **Initialize:** feasible $\mathbf{w}_i^{(0)}$ and $p_i^{(0)}$.
 - 2: **for** $n = 0, 1, 2, \dots, N-1$ **do**
 - 3: **Power update (convex):** Fix $\mathbf{w}_i^{(n)}$, solve the convex problem in p_i obtained from (13) to get $p_i^{(n+1)}$.
 - 4: **Beamformer update (SCA):** Fix $p_i^{(n+1)}$. Linearize all non-convex quadratic terms using (18) around $\mathbf{w}_i^{(n)}$, and solve the resulting convex problem to obtain $\mathbf{w}_i^{(n+1)}$.
 - 5: **Normalize:** $\mathbf{w}_i^{(n+1)} \leftarrow \mathbf{w}_i^{(n+1)} / \|\mathbf{w}_i^{(n+1)}\|_2, \forall i$.
 - 6: **if** $|\sum_i p_i^{(n+1)} - \sum_i p_i^{(n)}| \leq \varepsilon$ **then**
 - 7: **break**
 - 8: **end if**
 - 9: **end for**
 - 10: **Output:** $\mathbf{w}_i, p_i$.
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Subproblem 2: Beamforming Optimization with Fixed Power. In subproblem 2, the complex quadratic term is a non-convex term. The first-order Taylor expansion is the core method for approximating it with a convex lower bound.

At the iteration n , the quadratic term can be expressed by the first-order Taylor expansion as,

$$\begin{aligned} |\mathbf{a}^H(\theta_{i,t}) \mathbf{w}_i^{(n)}|^2 &\geq 2 \Re \left\{ \left(\mathbf{a}^H(\theta_{i,t}) \mathbf{w}_i^{(n)} \right)^* \mathbf{a}^H(\theta_{i,t}) \mathbf{w}_i^{(n)} \right\} \\ &\quad - |\mathbf{a}^H(\theta_{i,t}) \mathbf{w}_i^{(n)}|^2 = \mathbf{A} \end{aligned} \quad (14)$$

$$\begin{aligned} |\mathbf{a}^H(\theta_{i,t}) \mathbf{w}_k^{(n)}|^2 &\geq 2 \Re \left\{ \left(\mathbf{a}^H(\theta_{i,t}) \mathbf{w}_k^{(n)} \right)^* \mathbf{a}^H(\theta_{i,t}) \mathbf{w}_k^{(n)} \right\} \\ &\quad - |\mathbf{a}^H(\theta_{i,t}) \mathbf{w}_k^{(n)}|^2 = \mathbf{B} \end{aligned} \quad (15)$$

$$|\mathbf{h}_{i,j}^H \mathbf{w}_i^{(n)}|^2 \geq 2 \Re \left\{ \left(\mathbf{h}_{i,j}^H \mathbf{w}_i^{(n)} \right)^* \mathbf{h}_{i,j}^H \mathbf{w}_i^{(n)} \right\} - |\mathbf{h}_{i,j}^H \mathbf{w}_i^{(n)}|^2 = \mathbf{C} \quad (16)$$

$$|\mathbf{h}_{k,j}^H \mathbf{w}_k^{(n)}|^2 \geq 2 \Re \left\{ \left(\mathbf{h}_{k,j}^H \mathbf{w}_k^{(n)} \right)^* \mathbf{h}_{k,j}^H \mathbf{w}_k^{(n)} \right\} - |\mathbf{h}_{k,j}^H \mathbf{w}_k^{(n)}|^2 = \mathbf{D} \quad (17)$$

Thus, the beamforming optimization subproblem can be shown as follows,

$$\begin{aligned} \min_{\mathbf{w}_i} \quad & \sum_{i=1}^N p_i \\ \text{s.t.} \quad & \begin{cases} p_i \alpha_{i,t}^2 |\mathbf{a}(\theta_{i,t})|^2 \mathbf{A} \geq \gamma_r (\sum_{k=1, k \neq i}^N p_k \alpha_{k,t}^2 |\mathbf{a}(\theta_{i,t})|^2 \mathbf{B} + \sigma^2) \\ p_i \mathbf{C} \geq \gamma_c (\sum_{k=1, k \neq i}^N p_k \mathbf{D} + \sigma^2) \\ \|\mathbf{w}_i\|_2 = 1 \end{cases} \end{aligned} \quad (18)$$

At each AO iteration, the objective value $\sum_{i=1}^N p_i$ is non-increasing since each subproblem is solved optimally, and the SCA lower bounds are tight at the current iterate. Hence, the proposed AO-SCA algorithm converges to a stationary point of (10).

IV. SIMULATION RESULTS

A. Simulation Parameters

The radar task of the networked DFRC system is to detect 2 radar targets, and the communication task is to enable inter-aircraft communication of 3 aircraft. The settings of the remaining simulation parameters are shown in Table 1.

TABLE I
SIMULATION PARAMETERS OF THE NETWORKED DFRC SYSTEM

Parameters	Values
Maximum radiation power $P_{\max}$	100 W
Noise power σ^2	10^{-15} W
Radar detection SINR γ_r	15 dB
Communication SINR γ_c	8 dB

B. Simulation Results

Fig. 2 depicts the beamforming patterns of multiple aircraft in the networked ISAC system, where the red dotted line and the green dotted line respectively represent the spatial positions of target 1 and target 2. From the beam distribution characteristics of different aircraft shown in the figure, it can be clearly observed that the beam pattern designed by the proposed algorithm in this paper is consistent with the position of the detected target for all aircraft. It is not difficult to conclude that the proposed algorithm can effectively realize accurate beam pointing design for radar target detection, and ensure the consistency between the beam mainlobe and the target spatial position.

Fig. 3 compares the total radiation power of the networked DFRC system under different algorithms, where the abscissa is the typical algorithm schemes in the current DFRC field, and the ordinate is the total radiation power of the whole networked DFRC system. It can be seen from the power comparison results that the radar-centred algorithm obtains the lowest total radiation power among all algorithms, while the full power algorithm has the highest total radiation power without any power optimization design. The fundamental reason for the ultra-low power of the radar-centred algorithm is that it only optimizes the resource allocation with the radar detection performance as the sole objective, and completely ignores the communication performance constraints such as communication rate, link QoS, and user connection in the ISAC system, resulting in the loss of communication service function while saving power, which is difficult to meet the application requirements of DFRC system. For the full power algorithm, the system adopts the maximum radiation power for all aircraft without targeted power allocation, which not only causes serious energy waste but also easily leads to high inter-node interference and poor radio frequency stealth performance in the networked system. On the basis of satisfying radar performance and communication performance constraints, the proposed algorithm in this paper effectively reduces the total radiation power of the networked DFRC.

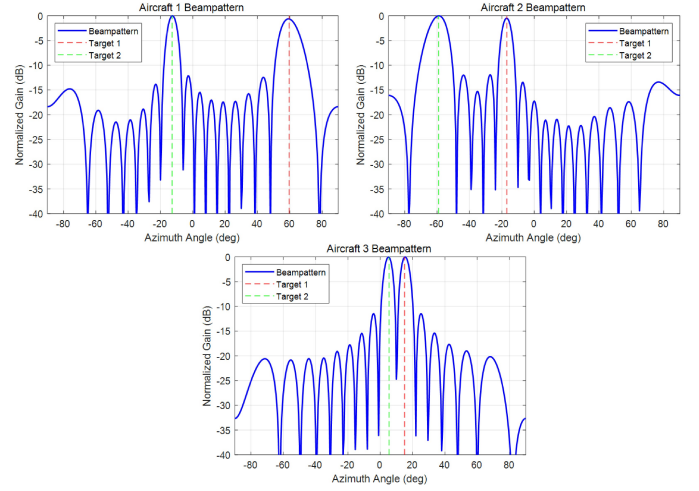


Fig. 2. Transmit beampattern of different aircraft in networked DFRC system.

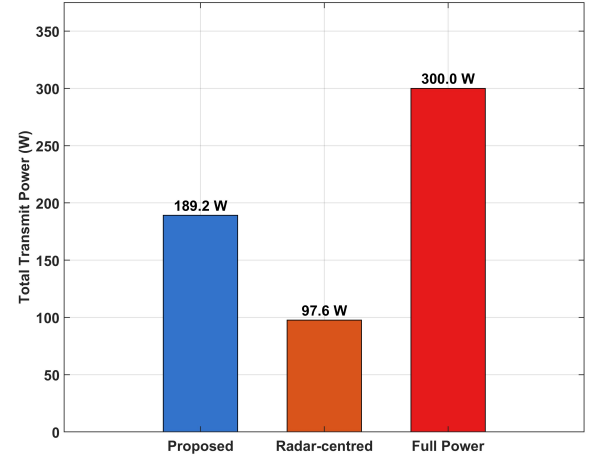


Fig. 3. Comparison of total radiation power in networked DFRC system.

V. CONCLUSION

This paper proposes a transmit beampattern optimization to improve the LPI performance of the networked DFRC system. The analytical expressions for radar SINR and communication SINR are derived, respectively. To optimize the total radiation power of the networked DFRC system, an optimization model is established under the constraints of satisfying the preset radar SINR threshold and preset communication SINR threshold. For the non-convex optimization model, an optimization solution is proposed by combining the AO method with the SCA method. The simulation results demonstrate that the proposed algorithm effectively reduces the total radiation power of the system, enhancing the LPI performance of the networked DFRC system.

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